TASK 1: PACKET IDENTIFICATION USING WIRESHARK

Roll Number: 2520030462

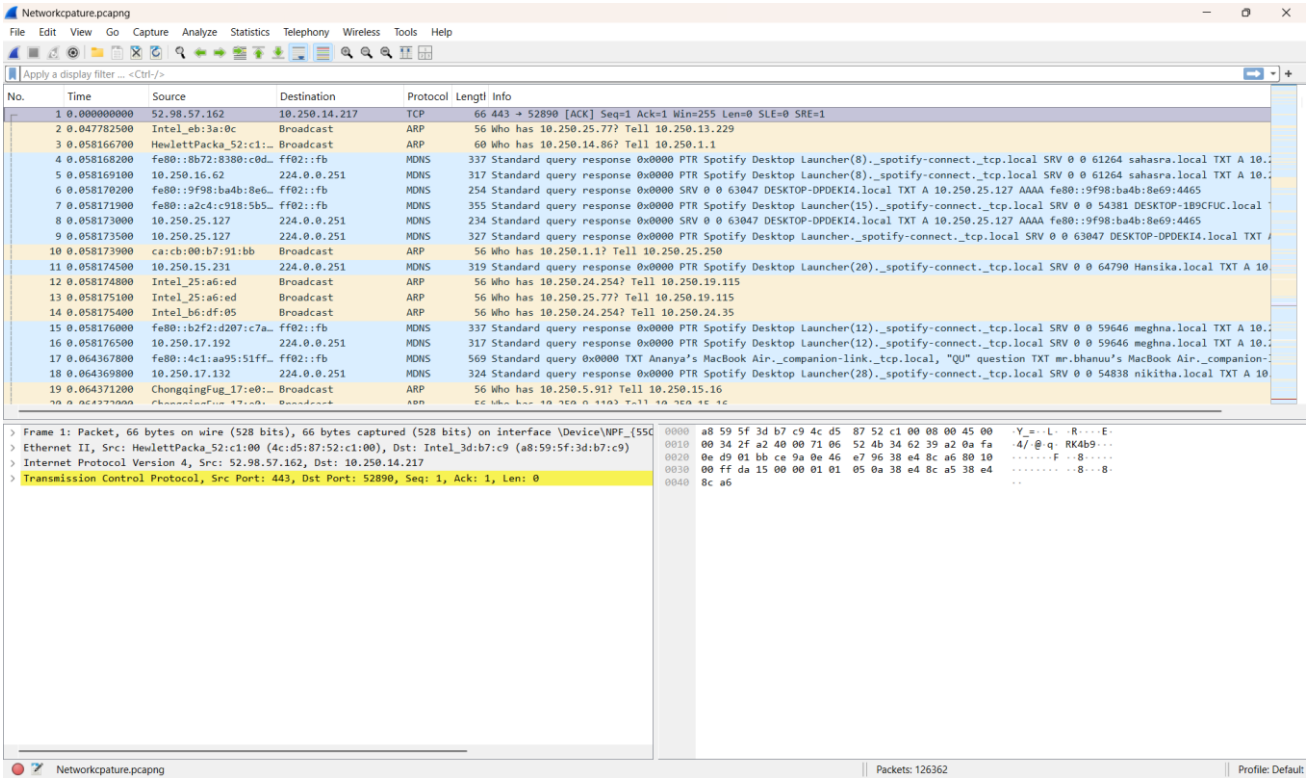
Name: B.Devi Charitha

1. Identify the Total Number of Packets Captured

Packet: A packet is a small unit of data transmitted over a computer network.

Procedure:

1. Open the Wireshark application on the computer.
2. Click on File → Open and select the previously created PCAP/PCAPNG file.
3. Click Open to load the captured network packets.
4. Wait until all the packets are displayed in the packet list.
5. Observe the bottom status bar of the Wireshark window.
6. Note the total number of packets captured.



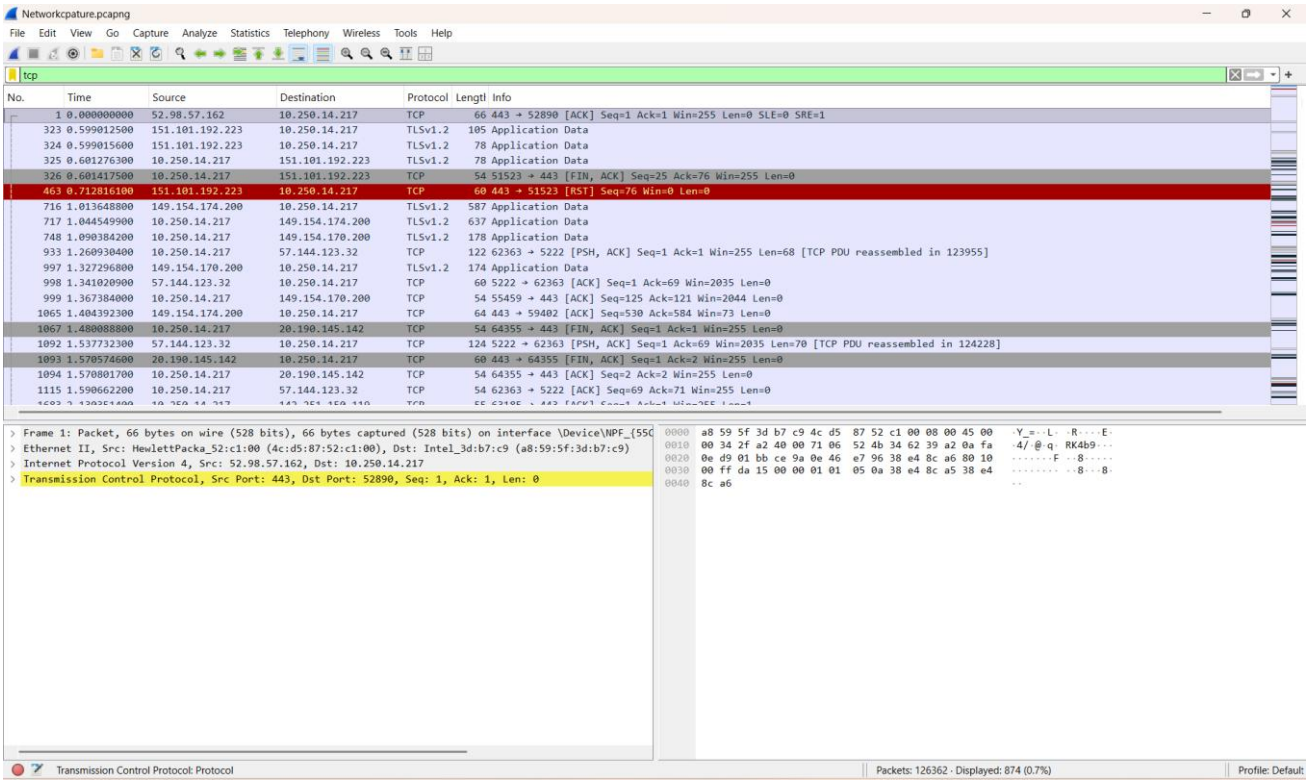
Result: The total number of packets captured was 126362.

2. Identify the Number of TCP Packets

TCP: TCP is a connection-oriented protocol that provides reliable data transmission between devices.

Procedure:

1. Open the PCAP file in Wireshark.
2. Locate the display filter bar at the top of the window.
3. Enter the filter tcp in the display filter bar.
4. Press the Enter key to apply the filter.
5. Wireshark displays only the TCP packets.
6. Observe the number of displayed packets at the bottom of the window.
7. Record the TCP packet count.



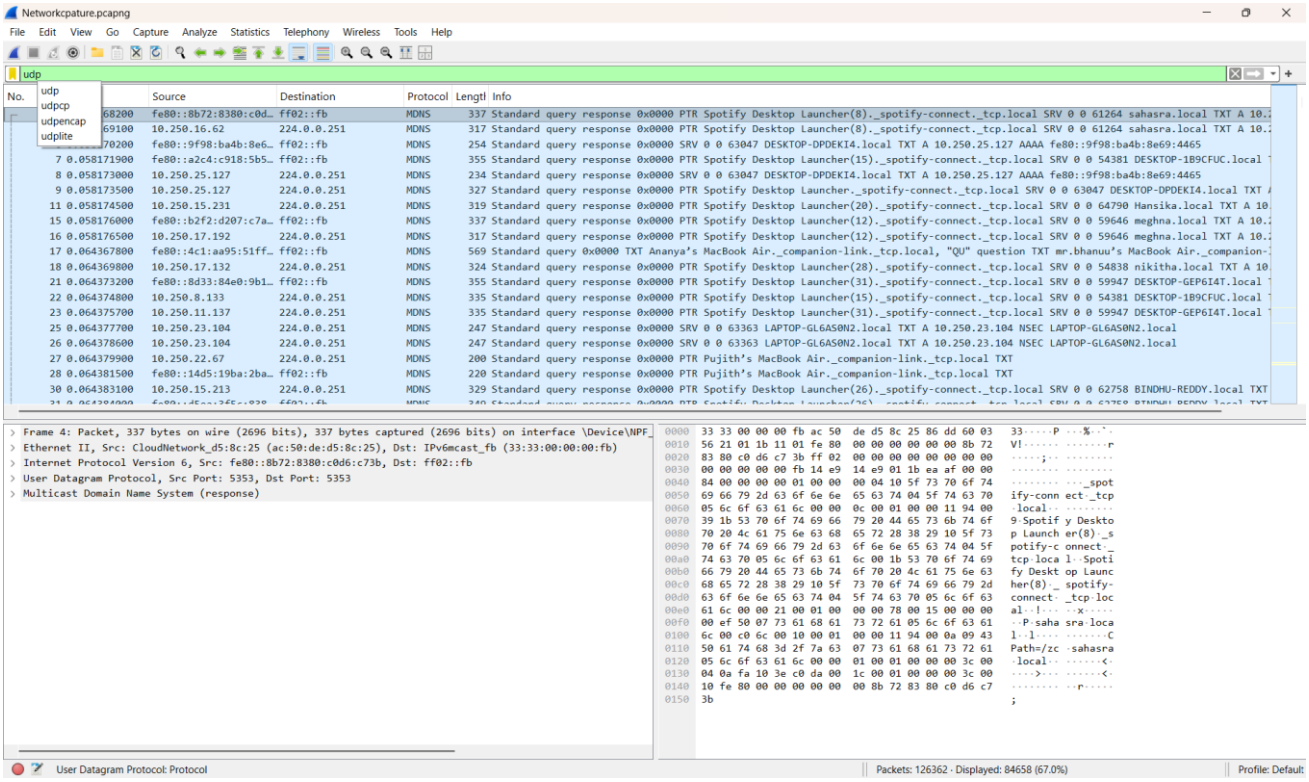
Result: The total number of TCP packets was 874.

3. Identify the Number of UDP Packets

UDP: UDP is a connectionless protocol that sends data quickly without guaranteeing delivery.

Procedure:

1. Open the same PCAP file in Wireshark.
2. Clear the previous display filter.
3. Enter the filter udp in the display filter bar.
4. Press Enter to apply the filter.
5. Wireshark displays only the UDP packets.
6. Observe the displayed packet count at the bottom of the window.
7. Record the UDP packet count.



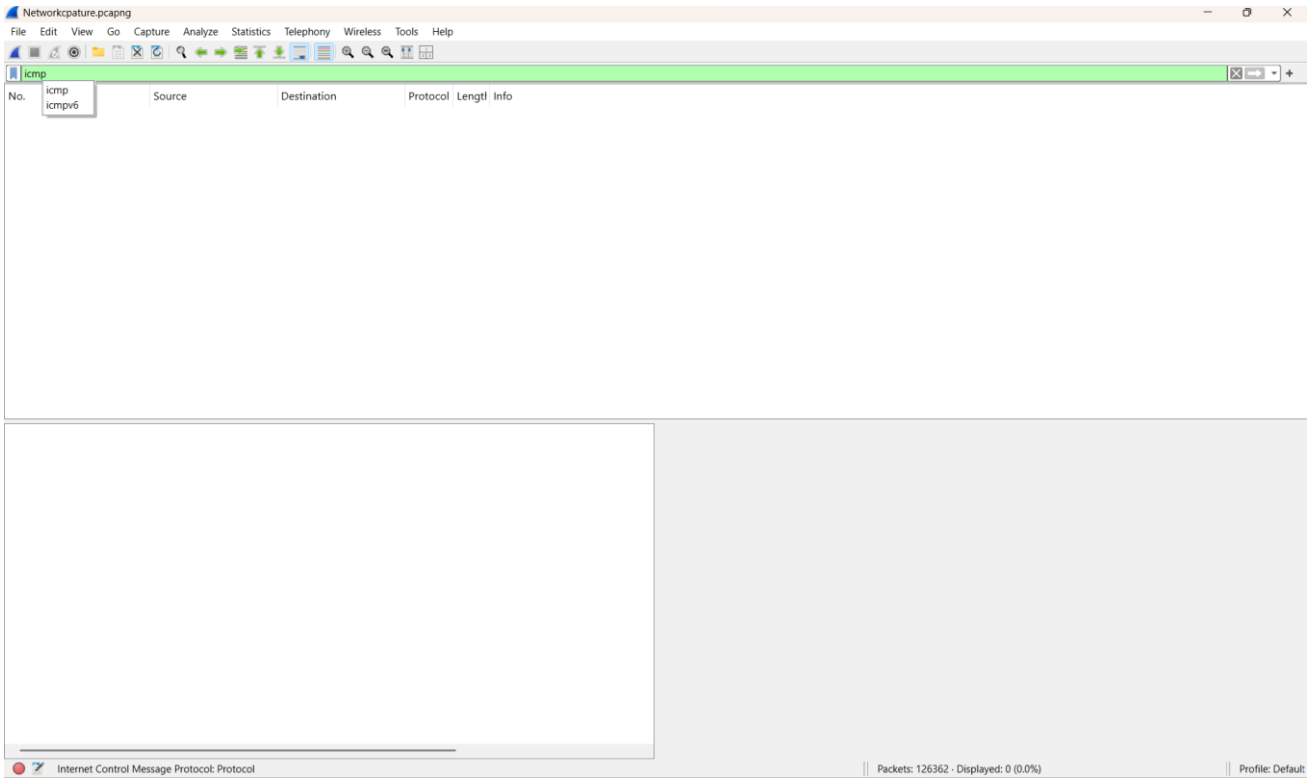
Result: The total number of UDP packets was 126362.

4. Identify the Number of ICMP Packets

ICMP: ICMP is a network protocol used to send error messages and diagnostic information. It is commonly used by the ping command.

Procedure:

1. Open the same PCAP file in Wireshark.
2. Clear the previous display filter.
3. Enter the filter icmp in the display filter bar.
4. Press Enter to apply the filter.
5. Wireshark displays only the ICMP packets.
6. Observe the number of displayed packets at the bottom of the window.
7. Record the ICMP packet count.



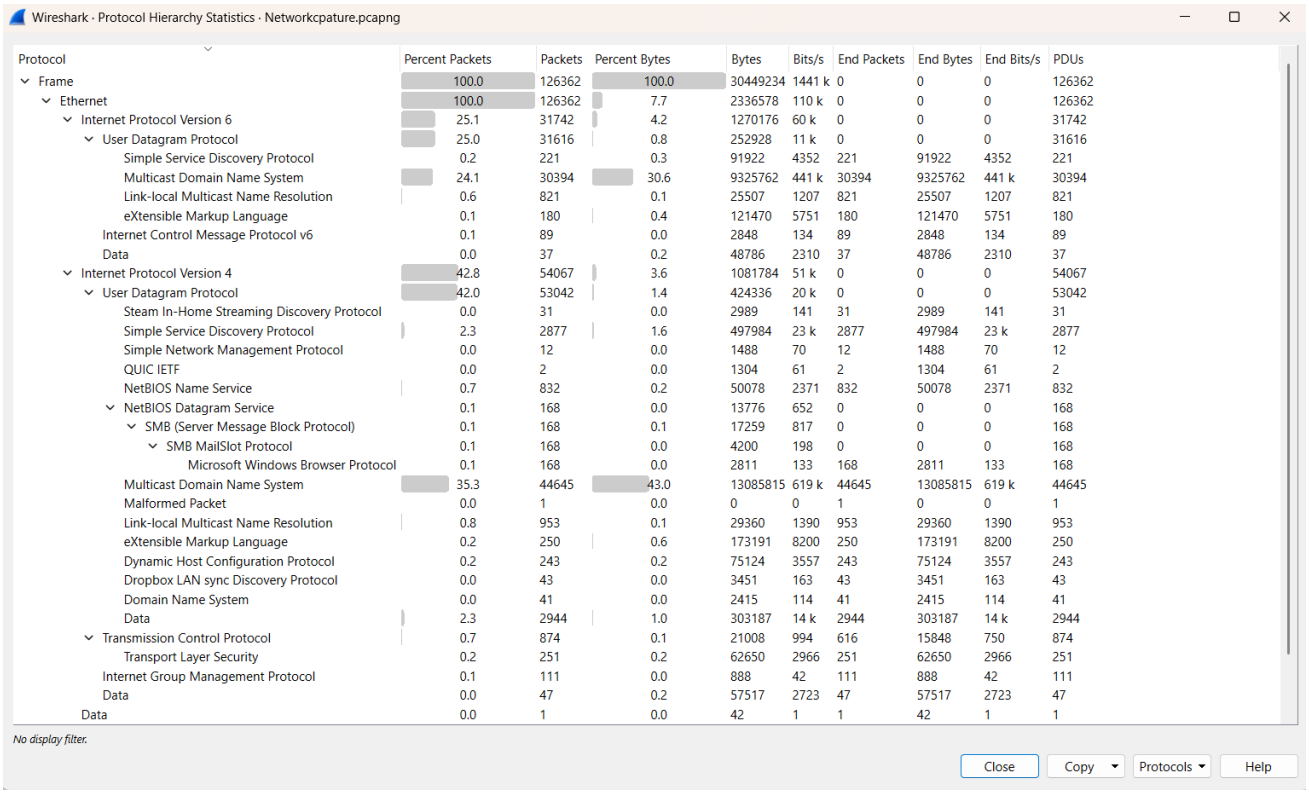
Result: No ICMP packets were found in the captured network traffic. The displayed packet count was 0.

5. Identify the Top Five Protocols Using Protocol Hierarchy

Protocol Hierarchy: Protocol Hierarchy is a Wireshark statistics feature that displays the protocols present in a capture along with their packet counts and percentages.

Procedure:

1. Open the PCAP file in Wireshark.
2. Remove any previously applied display filters.
3. Click on the Statistics menu in the menu bar.
4. Select Protocol Hierarchy from the dropdown menu.
5. A Protocol Hierarchy window will appear.
6. Observe the list of protocols displayed in the window.
7. Identify the top five protocols based on their packet counts.
8. Record the protocol names and their corresponding packet counts.



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	126362	100.0	30449234	1441 k	0	0	0	126362
Ethernet	100.0	126362	7.7	2336578	110 k	0	0	0	126362
Internet Protocol Version 6	25.1	31742	4.2	1270176	60 k	0	0	0	31742
User Datagram Protocol	25.0	31616	0.8	252928	11 k	0	0	0	31616
Simple Service Discovery Protocol	0.2	221	0.3	91922	4352	221	91922	4352	221
Multicast Domain Name System	24.1	30394	30.6	9325762	441 k	30394	9325762	441 k	30394
Link-local Multicast Name Resolution	0.6	821	0.1	25507	1207	821	25507	1207	821
eXtensible Markup Language	0.1	180	0.4	121470	5751	180	121470	5751	180
Internet Control Message Protocol v6	0.1	89	0.0	2848	134	89	2848	134	89
Data	0.0	37	0.2	48786	2310	37	48786	2310	37
Internet Protocol Version 4	42.8	54067	3.6	1081784	51 k	0	0	0	54067
User Datagram Protocol	42.0	53042	1.4	424336	20 k	0	0	0	53042
Steam In-Home Streaming Discovery Protocol	0.0	31	0.0	2989	141	31	2989	141	31
Simple Service Discovery Protocol	2.3	2877	1.6	497984	23 k	2877	497984	23 k	2877
Simple Network Management Protocol	0.0	12	0.0	1488	70	12	1488	70	12
QUIC IETF	0.0	2	0.0	1304	61	2	1304	61	2
NetBIOS Name Service	0.7	832	0.2	50078	2371	832	50078	2371	832
NetBIOS Datagram Service	0.1	168	0.0	13776	652	0	0	0	168
SMB (Server Message Block Protocol)	0.1	168	0.1	17259	817	0	0	0	168
SMB MailSlot Protocol	0.1	168	0.0	4200	198	0	0	0	168
Microsoft Windows Browser Protocol	0.1	168	0.0	2811	133	168	2811	133	168
Multicast Domain Name System	35.3	44645	43.0	13085815	619 k	44645	13085815	619 k	44645
Malformed Packet	0.0	1	0.0	0	0	1	0	0	1
Link-local Multicast Name Resolution	0.8	953	0.1	29360	1390	953	29360	1390	953
eXtensible Markup Language	0.2	250	0.6	173191	8200	250	173191	8200	250
Dynamic Host Configuration Protocol	0.2	243	0.2	75124	3557	243	75124	3557	243
Dropbox LAN sync Discovery Protocol	0.0	43	0.0	3451	163	43	3451	163	43
Domain Name System	0.0	41	0.0	2415	114	41	2415	114	41
Data	2.3	2944	1.0	303187	14 k	2944	303187	14 k	2944
Transmission Control Protocol	0.7	874	0.1	21008	994	616	15848	750	874
Transport Layer Security	0.2	251	0.2	62650	2966	251	62650	2966	251
Internet Group Management Protocol	0.1	111	0.0	888	42	111	888	42	111
Data	0.0	47	0.2	57517	2723	47	57517	2723	47
Data	0.0	1	0.0	42	1	1	42	1	1

Overall Conclusion

Wireshark was used to analyze the captured network traffic. The total number of packets was identified, and TCP, UDP, and ICMP packets were filtered using display filters. The Protocol Hierarchy feature was used to identify the top five protocols and understand the distribution of network traffic.